



Full Length Article

How to convert a 3D printer to a personal automated liquid handler for life science workflows

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ABSTRACT

Automated liquid handlers are fundamental in modern life science laboratories, yet their high costs and large footprints often limit accessibility for smaller labs. This study presents an innovative approach to decentralizing a liquid handling system by converting a low-cost 3D printer into a customizable and accurate liquid handler. The Personal Automated Liquid Handler (PALH) system, costing ~\$400, incorporates a single-channel pipet, custom 3D-printed components, and open-source software for personalized workflows, allowing researchers to build and modify the system for specific experimental needs. The PALH system was evaluated through common life science assays, including preparing real-time PCR samples, end-point PCR with novel pipet-based downstream purification, and genomic DNA extraction from peripheral whole blood. In real-time PCR experiments targeting the YWHAZ gene, the PALH system demonstrated comparable performance to manual preparation across DNA quantities (1 pg to 100 ng). For end-point PCR, the PALH successfully amplified and purified 204 bp and 406 bp amplicons from a pUC19 vector, yielding concentrations similar to manual methods (5.43 ± 0.85 ng/ μ L vs. 2.10 ± 0.16 ng/ μ L for 204 bp; 3.74 ± 2.13 ng/ μ L vs. 1.51 ± 0.15 ng/ μ L for 406 bp, respectively). In genomic DNA extraction from whole blood, the PALH system achieved comparable DNA yields to manual extraction (49.52 ± 3.13 ng/ μ L vs. 48.62 ± 5.9 ng/ μ L), although at higher purity (260/280 ratio of 1.83 ± 0.07 vs. 1.92 ± 0.03), although both are at acceptable ranges. The open-source nature of the PALH system hopefully encourages further community-driven improvements and protocol sharing, fostering innovation and collaboration within the scientific community. As laboratory automation advances, the PALH system could be crucial in democratizing access to high-quality automated liquid handling, particularly in resource-limited settings.

1. Introduction

Automated liquid handlers (ALH) have become essential tools for modern life science laboratories and play a pivotal role in how experiments are conducted. ALH robotic systems are designed to automate the precise handling of liquids, facilitating a wide range of applications such as sample preparation, DNA/RNA isolation, genetic or protein-based diagnostics, and high-throughput screening [1–5]. While each application requires specialized and specific automation workflows, they often address similar problems, reducing human errors and improving consistency, increasing throughput, reducing cost, reducing contamination risks, and improving ergonomics [3].

Despite the clear advantages of ALHs over manual workflows, current state-of-the-art liquid handling systems have several limitations. One of the primary drawbacks is their high upfront cost, which often prohibits smaller laboratories or research institutions from adopting

automation [6–8]. While the increased throughput capacity of ALHs can justify the high cost, it also results in systems with large footprints, posing a significant challenge for laboratories or testing sites with limited space. Furthermore, the increased throughput can strain the system's agility, locking it into a singular specialized workflow despite varying demands [9]. Therefore, in certain situations, utilizing reliable and flexible low-throughput systems may be more advantageous for personalized workflows in life science laboratories processes. This approach offers the benefits of automation without the burden of high number of sample volume requirements.

Additionally, the complexity of programming and operation compounds this rigidity of creating protocols. Most commercial ALHs are closed systems with proprietary software, making customization and integration with other laboratory equipment or workflows difficult [3, 8]. This lack of open-source flexibility can hinder innovation and limit the adaptability of the systems to specific experimental needs. However,

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high-quality manufacturing and low-upfront barrier-of-cost commercial automation systems with open-sourced protocol building are continuously being introduced to the market, most notably and recently by Opentrons, but still start at around USD 15,000 with additional costs for modules [10]. Perhaps, Opentrons system is one of the only commercialized open-source ALHs with extensive library of labware, unique functions for different workflows and broader flexibility in programming and scripting. Other automation systems such as *OpenWorkstation* [11], *Syringe pump set* [12], *EvoBot* [13], *Automated liquid handler from a 3D printer* [14], and *Liquid handling Lego robots* [15] are great examples of open-source DIY systems with substantially lower cost. A comparison of notable select commercial and open-source liquid handlers is shown in Table 1.

Open-source automation platforms have played a crucial role adaptation of the liquid handling systems for enabling researchers to implement custom solutions. These platforms have facilitated the creation of custom modular systems, such as those described in the works of Eggert et al. which have demonstrated improved efficiency and reproducibility in tasks ranging from sample preparation to data analysis for applications of production and characterization of tailorable gelatin methacryloyl (GelMA)-based hydrogels or 3D cell culture models [16–19].

Automated liquid handlers typically employ a common mechanism based on a 3-axis XYZ gantry robotic system [20]. This design allows for precise movement and positioning of the liquid dispensing components across a defined workspace. The core of the liquid handling process involves liquid displacement, usually achieved through pipet or syringe mechanisms. Many systems utilize disposable tips to prevent cross-contamination between samples. The workspace, often called the deck, is designed to accommodate various modules such as heating elements, magnetic separators, and plate or tube holders, enabling a wide range of experimental protocols.

Interestingly, the fundamental design of liquid handlers shares striking similarities with 3D printers, particularly in their use of 3-axis gantry mechanisms. The evolution of personalized 3D printers and the open-source movement has led to many accessible, customizable, and cost-effective 3D printing solutions [21]. Typically, 3D printers employ stepper motors to control the movement of the gantry on each axis. These motors provide precise, incremental motion by rotating a specific number of steps in response to electrical pulses controllable by a user. This level of control enables the printer to achieve precise spatial resolution, often in the range of micrometers to millimeters [22]. Central to the mechanism of 3D printers is the use of G-code, a standardized programming language for computer numerical control (CNC) machines. G-code controls specific motors through preprogrammed software codes embedded in the system. The G-code in typical 3D printer software includes movement, extrusion, temperature control, and fan control commands [23].

This trend of decentralizing 3D printer development has inspired the adaptation of 3D printer devices for liquid handling applications in research settings, notably by Kopyl et al., Barthels et al., and Li et al., for educational purposes [14,24,25]. Not only for automated liquid handling, 3D printers have also been repurposed for various other applications in life science workflows [26]. For instance, Rauh et al. converted the Ender-3 3D printer to a versatile multi-layer dip coater for thin film deposition [27] or Cordova et al. developed the Enderstruder, 3D printer syringe extruder for precise material deposition for bioprinting applications [28] or Ponzetti et al. introduced the HistoEnder, a 3D printer-based histological slide autostainer [29]. Broader use cases of 3D printers in laboratory settings, such as applications in creating custom labware, molecular models, and even functional devices for chemical synthesis and analysis, are also noteworthy [26]. These examples illustrate how repurposed 3D printers can serve as versatile, low-cost alternatives to specialized commercial equipment across various life science workflows.

In this article, we present a single-channel personal automated liquid

Table 1

Comparison of commercially available and reported open-source ALH systems.

Category	Instrument / Project	Pros	Cons
Commercial	Tecan EVO 75 [31]	Flexible, modular design, scalable	High initial cost, complex programming
	Hamilton Microlab STAR [32]	Precise, reliable, wide volume range	Large footprint, high maintenance
	Agilent Bravo [33]	Fast, robust, good integration with other Agilent instruments	Limited deck space, not modular
	INTEGRA VIAFLO [34]	Easy to use, relatively affordable, multiple tip options	Limited volume range, limited high throughput applications
	Eppendorf epMotion [35]	Reliable, user-friendly concrete protocols	Slower than some high-end systems, limited deck space
	Revvity Janus Workstations [36]	Can handle a variety of labware formats, flexible liquid handling mechanism	High initial cost, complex setup, requires more programming expertise for protocol development
Open-Source	Beckman Coulter Biomek i-Series [37]	Flexible, good for complex workflows	High cost, large footprint
	Thermo Scientific Versette [38]	Compact design, easy to use, good for medium throughput applications	Limited deck space, not for high-end applications
	OpenWorkstation [11]	Modular, customizable, storage space, and has a temperature dock	Requires programming knowledge to operate, limited opensource hardware adoption, a stand-alone pipet with limited volume change
	Ender-3 3D printer kit [12]	3D printer block for low cost and use readily available parts	Limited functionality compared to other ALH, a syringe pipetting module with no plate loading mechanism
	Computer-controlled linear motion pipet [39]	Open-source, cost-friendly, modified pipet	Requires assembly expertise and a stand-alone pipet with limited volume change
	EvoBot [13]	Open-source, modular, multiple pipet tips, and designed for scientific experiments	Limited information, relatively high error rate
	Automated Liquid Handler from a 3D Printer [14]	Utilizes 3D printer for a low-cost option and low-barrier assembly	Requires a stand-alone pipet with limited volume change
	FINDUS [40]	Open-source, 3D printable, and specifically designed for life sciences	Limited information, large footprint, complex assembly, no fixed dock
	Opentrons Flex/OT2 [10]	Modular, customizable, relatively affordable compared to other commercial systems and flexible throughput	Relatively large footprint

(continued on next page)

Table 1 (continued)

Category	Instrument / Project	Pros	Cons
	OpenLH [24]	Accurate, repetitive, easy to operate, affordable, small footprint	Single pipet with limited volume capacity
	Liquid-handling Lego robots [15]	Can perform complex mixing protocols using off-the-shelf Lego parts, relatively inexpensive	Requires building and setup guidance, reproducibility, and long-term usage may be limited due to plastic parts

handler (PALH) that utilizes the principles of accessibility, customization, and open-source development for democratizing liquid handling systems further using Ender-3 3D printer (Fig. 1) [30]. With self-programmable software, researchers can use PALH to displace liquid, execute magnetic bead-based purification, and heat pipet tip. PALH is designed to be easily assembled, allowing researchers to build and modify the system according to their requirements. Additionally, PALH possesses the capability to 3D print its own parts and deck components, enhancing its versatility and reducing dependency on external suppliers. The open-source nature of the PALH extends to protocol development, enabling researchers to create, share, and optimize assay protocols freely. This approach reduces costs and fosters innovation and collaboration within the scientific community.

2. Results and discussion

2.1. Hardware fabrication and design

This study evaluates the feasibility of repurposing 3D printers as cost-effective personal automated liquid handling systems for life science assay workflows. The PALH system comprised three major components: 1) 3D printer XYZ gantry rails frame enclosed with electronics for precise pipet movement, 2) custom pipet module for liquid handling within pipet tips, and 3) custom deck base to securely hold consumable modules (pipet tips, tubes, well plates), module a magnet holder, pipet

heater, and disposable tip waste ejector container and trigger. A cost-effectiveness analysis was conducted based on the assembly components' respective unit prices (Table 2). Including additional mechanical components, printing consumable filaments, and assembly parts, the total base cost of the PALH system was USD 400.55. The addition of magnetic bead-based purification capacity increased the base cost by USD 4.08, while additional wiring to simplify the assembly or having an enclosure for the PALH increased the price by USD 216.99.

The fused deposition modeling (FDM) 3D printer was selected for the PALH system due to its decreasing cost, design flexibility, and precision control of its XYZ gantry system [41]. Among various models, the Creality Ender-3 was selected as the optimal platform for this study due to its competitive cost-effectiveness and modular architecture. (Figure S1) [42]. The Ender-3's open-source design facilitates the modification of components, firmware, and software, enabling efficient hardware updates and protocol development.

The assembled Ender-3 3D printer was initially employed to fabricate custom, non-manufactured components for the PALH. However, the printer's firmware configuration needed to be modified to optimize the available deck space and accommodate additional modules. The default firmware imposes safety-related limitations on the printable bed size; hence, the soft limits were extended by 3 mm along both the X and Y axes through alterations to the *configuration.h* header file before firmware build (the firmware files are publicly accessible via Zenodo) [43]. Two 3D print files were printed for the PALH assembly: 1) *deck_components.gcode*, contains the deck base model, and 2) *assembly_components.gcode*, contains all additional assembly components (Table S1; Both the 3D models and their corresponding print files are publicly available via Zenodo) [43]. The printing bed mat is secured to the printer bed with binder clips at its four corners during printing. The 3D model for the deck component includes cutouts at the corners to accommodate these clips (Figure S2), minimizing interference with the printing process.

After all components were printed, the 3D printer's extruder was replaced with a liquid-displacing pipet cone assembly instead of fitting a conventional manual handheld pipet into the printer. Thermo Scientific's single channel E1 ClipTip cone assembly pipet with a 2–125 μ L volume range was assembled into the 3D printer to replace the extruder in the X-axis rail. Depending on the workflow, the pipet cone can be

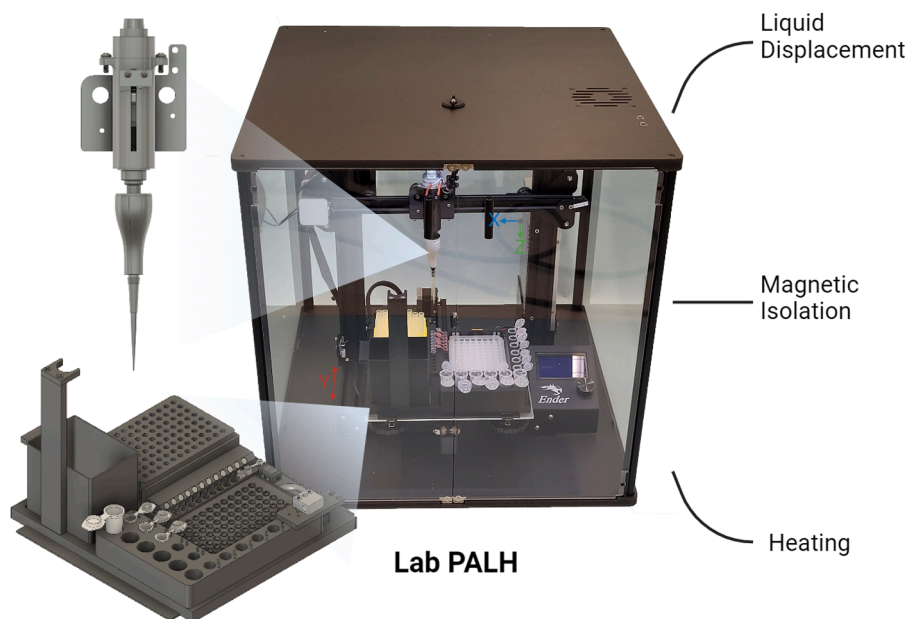


Fig. 1. A customizable, open-source single-pipet liquid handling system, PALH, was built by modifying an Ender-3 3D printer with a custom pipet module (left-top) and custom deck base module (left-bottom). The system can carry simple liquid handling processes of liquid displacement using the pipet module, pipet-based magnetic purification using the magnetic stand on the custom deck base module, and pipet heating using the extruder heater on the custom deck base module.

Table 2
Cost analysis of components used for assembling the PALH system.

Assembly components	Catalog No	Manufacturer/Supplier	Quantity	Unit Price
Creality Ender-3 3D printer	Ender-3 V4.2.7	Creality	1	\$168.99
E1 ClipTip, tip cone assembly, plastic tip fitting, single channel, 125µL	TS 2215590N	Thermo Scientific/ Pipet Supplies	1	\$127.00
Can-Stack stepper motor linear actuator (G4 25,000 25mm)	25,443-12	Haydon Kerk Pittman	1	\$91.73
Micro limit switch (5A 125V-250V 3 Pins, Type 4)	MXR-PL-YBKG	MXRS/Amazon	7	\$2.33
2-56×1/2" Pan head Phillips screw (for fixing pipet limit switch)	91772A081	McMaster-Carr	2	\$0.13
2-56×3/4" Phillips flat head screws (for fixing limit switches on deck)	91500A304	McMaster-Carr	3	\$0.25
6-32×3/8" Phillips head machine screw (for fixing actuator to pipet casing)	91772A146	McMaster-Carr	2	\$0.13
6-32×1/2" Phillips flat head screws (for fixing pipet trigger on deck, top)	90273A148	McMaster-Carr	2	\$0.10
6-32×1" Phillips flat head screws (for fixing pipet trigger on deck, bottom)	90273A153	McMaster-Carr	2	\$0.13
Ender 1.75mm PLA 3D printing filament 1kg	Ender 1.75mm PLA	Creality	0.5	\$10.00
Neodymium disc magnet (optional)	D42	K&J Magnetics	12	\$4.08
3D printer extension cable (optional)	B093QQFBZS	Lzjsee/Amazon	1	\$17.99
3D printer enclosure (optional)	B09QGCR548	FNATR/Amazon	1	\$199.00
			Total	\$400.78 + \$221.07 (optional)

easily interchangeable with different volume ranges (0.5–12.5µL, 10–300µL, or 15–1250µL). The E1 ClipTip's interlocking clipping technology, which uses custom Thermo Fisher lockable disposable tips, provided a secure seal without requiring significant force, unlike the friction sealing systems, making it ideal for PALH and reducing reliance on strong z-axis motors [44,45]. The pipet component encased the E1 ClipTip cone, limit switch, limit switch trigger, linear actuator, and linear actuator trigger (Fig. 2B). The 3D-printed linear actuator trigger connected the stepper motor linear actuator with the E1 ClipTip Cone

via the plunger and controlled the air and liquid volume during displacement. At the start of every protocol or run, homing positions for all the motors must be identified. Hence, the pipet encasing has a placeholder for positioning a homing limit switch and is triggered by the limit switch trigger. A 3D-printed pipet cone head is aligned with the pipet cone, which is pushed down with a pipet tip ejector trigger on the deck space to eject the tips from the pipet cone. All the ejected waste tips are collected in the waste container (Fig. 2A–2C).

Once the pipet component was assembled, the extruder module from the 3D printer board was replaced with the pipet component along the X-axis rail, and the Z-axis rail homing limit switch was displaced (102.22 mm above the original positioning) to ensure the pipet cone head does not crash into any modules. Securing the deck base onto the bed using binder clips once the printing mat was removed was important; therefore, the four binder clips were manually moved to clamp the deck space after the print job was done (Fig. 7A–B). The 3D printer's built-in electronics conveniently facilitated the movement of the pipet actuator and the pipet homing, eliminating the need for electronics reassembly. However, it was important to rewire the connection between the board and the motors, so the phasing of the motor matches the power output, as illustrated in (Fig. 7C). The pipet actuator's homing limit switch was also rewired to the extruder's thermal sensor to detect the homing trigger. However, this convenience necessitated manually fine-tuning the potentiometer in the mainboard from the extruder stepper (V_{ref}) to 0.15–0.06 V_{DC} . This adjustment ensured the actuator did not overheat while providing enough power to move the motor.

The 3D-printed deck space encompassed all necessary modules for the PALH, including the pipet tip ejector, the E1 ClipTip tips, a plate, and tubes. Four different types of module placeholders can be positioned here: a single 96/384-well plate, 12 slots of Type 1 1.5 mL/2.0 mL tubes, 6 slots of Type 2 5.0 mL tubes, and 24 slots of Type 3 0.1 mL/0.2 mL tubes. The positioning of each tube location is specified in Fig. 2A, referenced in the software during protocol script design.

Many life science workflows require magnetic bead-based purification [46–48]. To support such processes, a novel pipet-based pelleting mechanism was used in the PALH, where a magnetic field is applied outside the pipet when handling magnetic beads. Such a process is possible with a magnet module holding a series of simple neodymium permanent magnets (2.30–3.71 lbs pull force) positioned at the first row (well locations 1–12) of the Type 3 tube placeholder on the deck (Fig. 2C) [49]. Here, the pipet tip is positioned in close proximity to the stationary magnetic disks to pellet the beads to the side of the pipet tip. The magnet holder module positions 12 magnetic disks at 45° to pellet magnetic beads, where the strongest magnetic field is experienced at the edge of the magnetic disk (Figure S3). By placing the pipet tip in close proximity to or away from the stationary magnetic disks, the system can simulate the "on" and "off" modes of a magnet.

For instance, the pipet-based magnetic bead purification process involves a series of steps to ensure efficient nucleic acid isolation (Fig. 3). Initially, the magnetic bead mixture with the sample with a biomarker of interest, such as nucleic acid, is slowly aspirated into the pipet tip and pelleted when aligned with an external magnetic disk. This slow aspiration is crucial for ensuring complete bead pelleting before the supernatant is carefully dispensed back into the input well. The dispensing speed is deliberately slow to maintain the integrity of the bead pellet and minimize bead loss by preventing disruption from the liquid-air interface. Following dispensation, the tip is rapidly withdrawn from the magnetic field along the Z-axis before moving to a new wash buffer well. The washing step can be performed either in the presence of a magnetic field for gentle washing with compact bead pellets or without a magnetic field for more thorough mixing and stringent washing. Finally, the targeted nucleic acids are eluted from the beads in an elution buffer without a magnetic field, allowing complete bead resuspension. The eluted material is then separated from the beads using the magnetic field for final product collection. It is worth noting that while this novel pipet tip magnetic purification procedure does not require physical

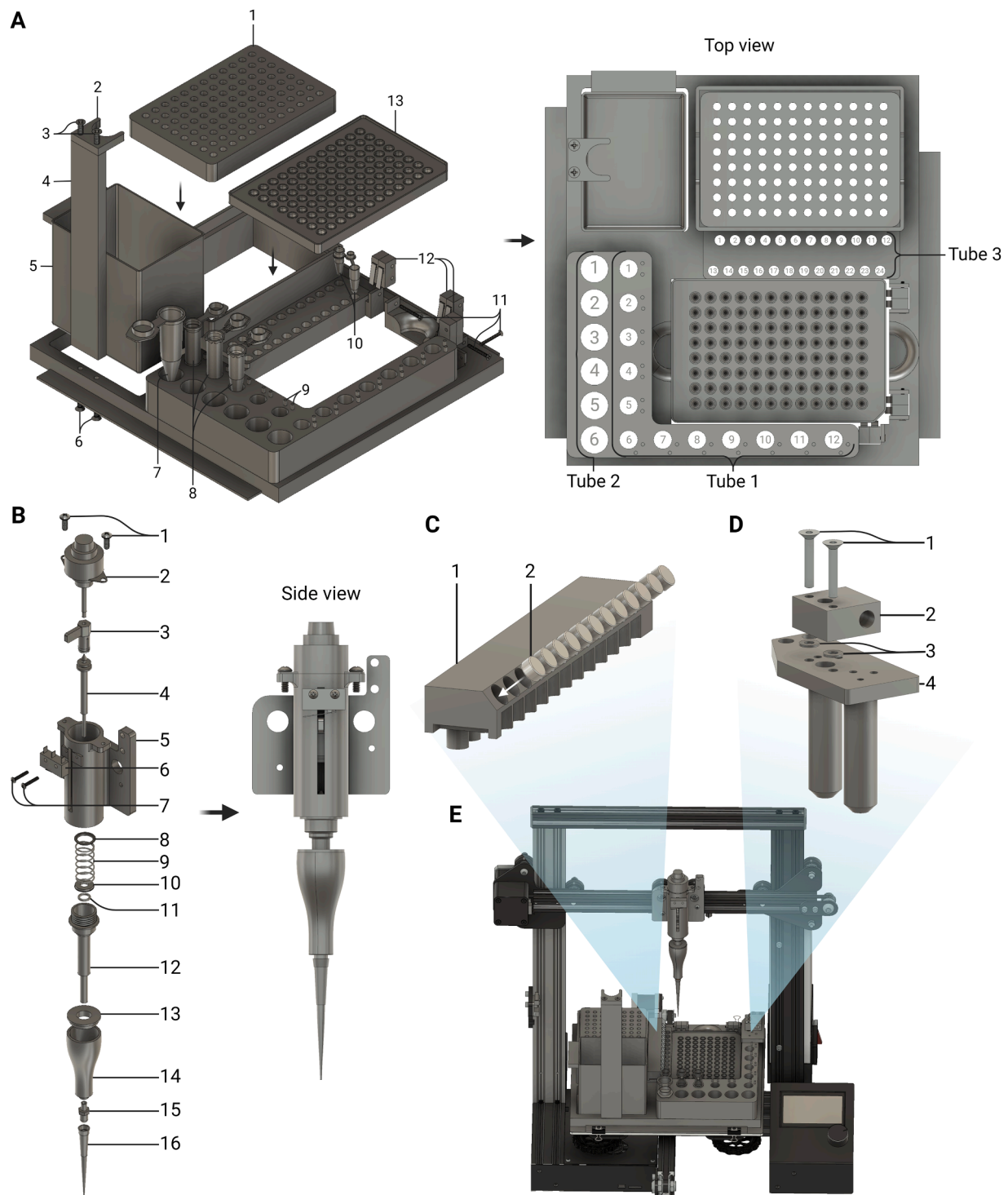


Fig. 2. PALH system's components. A) Deck base space (3D-printed) for holding modules: 1. Disposable ClipTip pipet tip holder. 2. Pipet tip ejector trigger (3D-printed), 3. Two 6-32×1/2" screws for securing pipet tip ejector trig to ejector neck top. 4. Pipet tip ejector neck (3D-printed), 5. Waste tip container, 6. Two 6-32×1/2" screws for securing pipet tip ejector trigger to ejector neck bottom (3D-printed), 7. Placeholder for 5.0 mL tubes ('Tube 2' type), 8. Placeholder for 1.5 mL/2.0 mL tubes ('Tube 1' type), 9. Pins for guiding and securing tube cap joint, 10. Placeholder for 0.1 mL/0.2 mL rimmed and rimless tubes ('Tube 3' type), 11. Three 2-56×3/4" screws for securing limit switches around the plate, 12. Six repurposed limit switches for positioning the plates at the corner of the plate placeholder, 13. 96/384 well plate. B) Pipet assembly in replacement of the extruder: 1. Two 6-32×3/8" screws to secure the stepper motor actuator to the pipet casing, 2. Stepper motor linear actuator, 3. Homing limit switch trigger (3D-printed), 4. Plunger, 5. Pipet casing (3D-printed), 6. Homing limit switch, 7. Two 2-56×1/2" screws to secure the limit switch to the pipet casing, 8. Top spring retainer, 9. Spring for plunger, 10. Bottom spring retainer, 11. O-ring seal, 12. Plunger guide, 13. Pipet cone head (3D-printed), 14. Pipet cone, 15. Pipet tip ejector, 16. Disposable ClipTip. C) Magnet holder module positioned at the first row (positions 1–12) of Tube 3 placeholder: 1. Magnet holder (3D-printed), 2. Magnet disks. D) Heater module for thermally treating pipet tips: 1. Extruder screws, 2. Repurposed extruder heater block, 3. Insulator washer between the heater block and the holder, 4. The heater block holder is positioned at the end of the Type 2 tube placeholder (positions 11–12; 3D-printed).

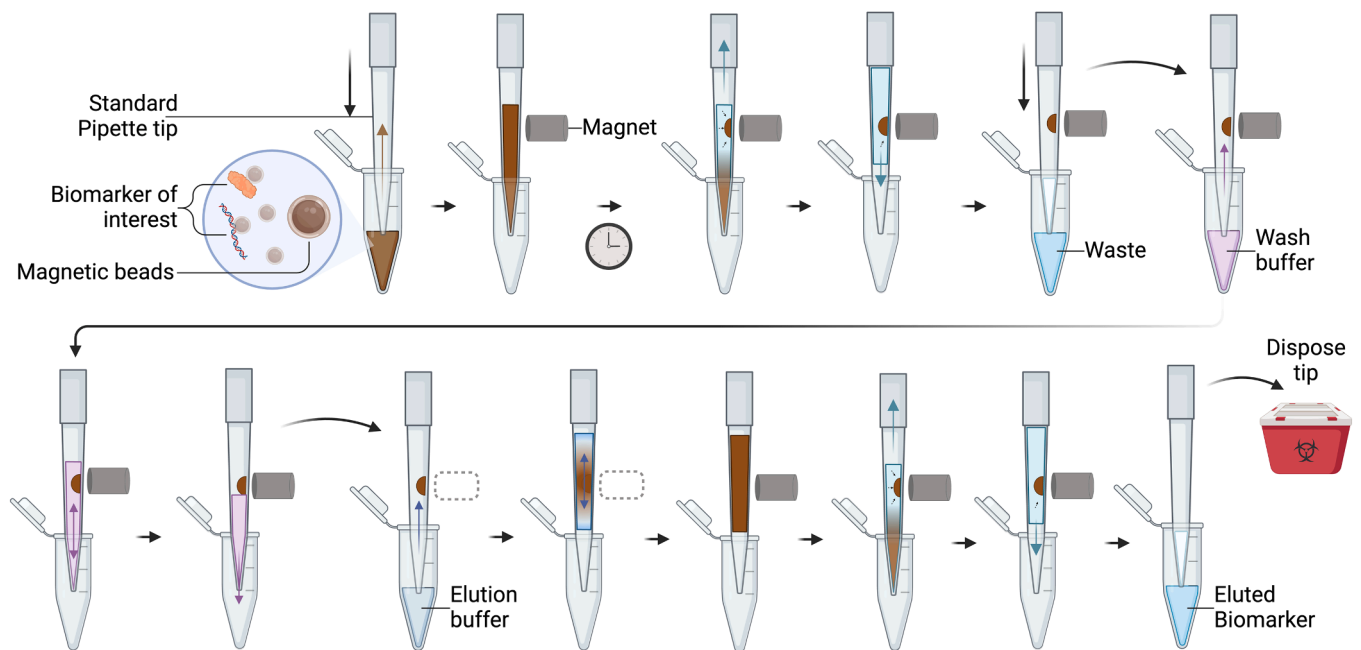


Fig. 3. Schematic illustration of pipet-based magnetic purification workflow with bead pelleting during washing procedure. A standard pipet tip aspirates a sample containing magnetic beads and biomarkers of interest. By positioning the external magnet, the magnetic beads are pelleted against the inside of the pipet tip wall, allowing the separation of waste supernatant. The pelleted beads undergo washing steps with a wash buffer while maintaining magnetic capture. Finally, the biomarker is eluted from the magnetic beads using an elution buffer in the absence of a magnetic field, followed by a final magnetic separation to isolate the purified biomarker before the pipet tip is disposed of.

activation or deactivation of the magnet, the rapid movement of the pipet tip towards and away from the magnet may potentially compromise the compactness of the pelleted beads, risking bead loss.

In certain protocols, these magnetic beads are dried to evaporate any wash buffer using heat or to agitate the magnetic beads during elution step at a heightened temperature to increase nucleic acid yield [50,51]. Hence, the PALH system incorporated a heating module to compensate for such workflows (Fig. 2D). Here, the extruder nozzle heater's power wire was reconnected to the original hotbed power source port on the mainboard while the thermistor wire was reconnected to the original hotbed thermistor port (reconfigured board connections are illustrated in Fig. 8).

2.2. Firmware and protocol builder program structure

There are two modified versions of the open-sourced Marlin firmware, one for printing the deck base, as previously mentioned, and another for the PALH. The only modification in the printing firmware is the change in the bed size limit. The PALH firmware, on the other hand, has multiple different modifications. For example, the PC5 pin from the mainboard is reassigned as the limit switch for homing the pipet actuator and the E stepper port for controlling the pipet stepper motor actuator as parameter 'V'; this is referenced in the G-code lines when generating protocol scripts. Definitions such as direction of homing and steps for a unit of volume (μL) are also reflected in the firmware. The ideal steps required for the selected stepper motor actuator for displacing volume were determined as 165 steps/ μL when displacing 100 μL of water at 20 $\mu\text{L}/\text{second}$ (Table S2). The volume displacement accuracy was tested at various volumes. The complete modifications of the firmware can be accessed via Zenodo [43]. The newly built modified firmware (.bin) was copied to a microSD card for running protocols on the PALH system.

The G-code lines control the PALH system motors. Hence, a custom Python program (*gcode_generator_v1.2.py*) was developed to simplify generating G-code lines. This program reflects the definitions of all modules (Tube 1–3 and 96/384 well plates) and the functions to be used

for creating protocols [43]. The X and Y-axis coordinates (the distance in mm from the homing positions) of the module locations are defined in Table S3, and the Z-axis bottom of the well and the top of the well opening are defined in Table S4. These coordinates are important when referencing a specific module well or tube locations from which to source or direct the pipets.

When handling liquid, bubbles need to be prevented during displacement, which can interfere with volume displacement accuracy. Bubbles often form when displacing air into a liquid or if the velocity of the liquid displacement is inappropriate (too fast); therefore, the correlation of height represented in Z-axis coordinates and the volume of liquid was important to determine. The volume-to-height correlation depends on the geometry and volume of the modules; thus, the coordinates of the pipet tip contact at the liquid's surface were determined, and the best-fit line with R^2 above 0.99 was referenced from the Gcode generator program when the dispensing height was required to avoid bubble formation where the pipet tip is 1–3 mm above the liquid surface during dispense depending on the module (Fig. 4). The volume-to-height correlation was split into upper and lower ranges since most modules tapered towards the bottom of the well or tube, where the slope of the best-fit-line differed. This splitting approach was necessary because the tapering geometry of the different modules resulted in a non-linear relationship between volume and height, with a steeper slope in the lower portion. Hence, the best fit for each module was created in two separate volume ranges, except for the 384-deep well plate – which has a more linear relationship between volume and Z-coordinates [52]. The separate ranges helped PALH functions to more accurately model the volume-to-height relationship across the entire module, accounting for the change in cross-sectional area due to tapering.

A combination of unique functions was written to facilitate the ease of protocol development, and the highlights of some functions are referenced here (Fig. 5). The complete scripting source codes and example codes can be accessed via Zenodo [43]. For example, to simplify creating new plate modules, the *well_creator* function was written. In creating new plate modules the following parameters were used: the number of rows, columns, spacing between wells (in mm), the well

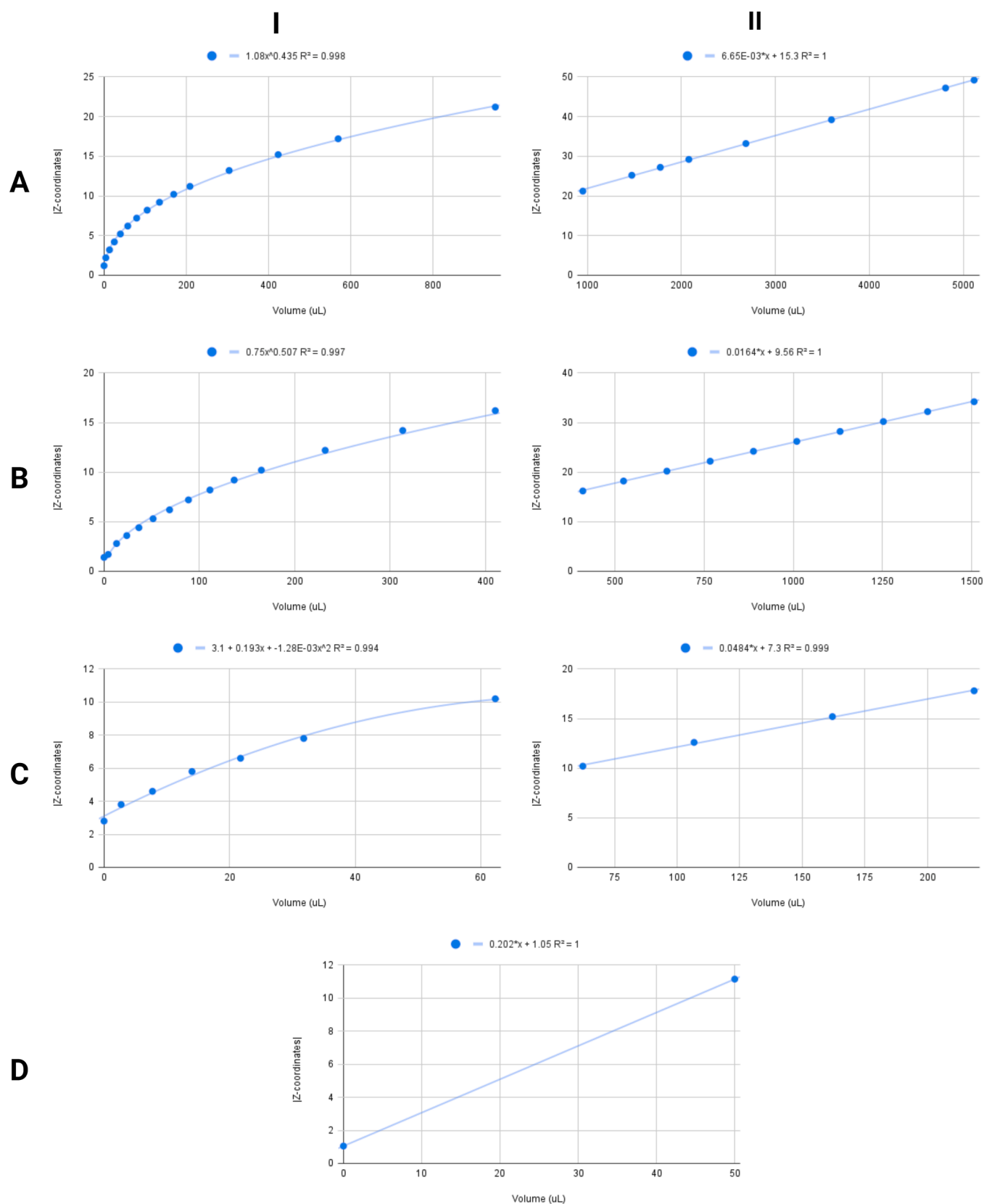


Fig. 4. The volume height (μL) and the absolute Z-axis coordinate correlations of different modules with the best-fit line equation and its corresponding R^2 . The lower range (I) and the upper range (II) of modules A) 5.0 mL tube, B) 1.5 mL tube, C) 0.2 mL tube/ 96-well plate, and D) 384-well plate.

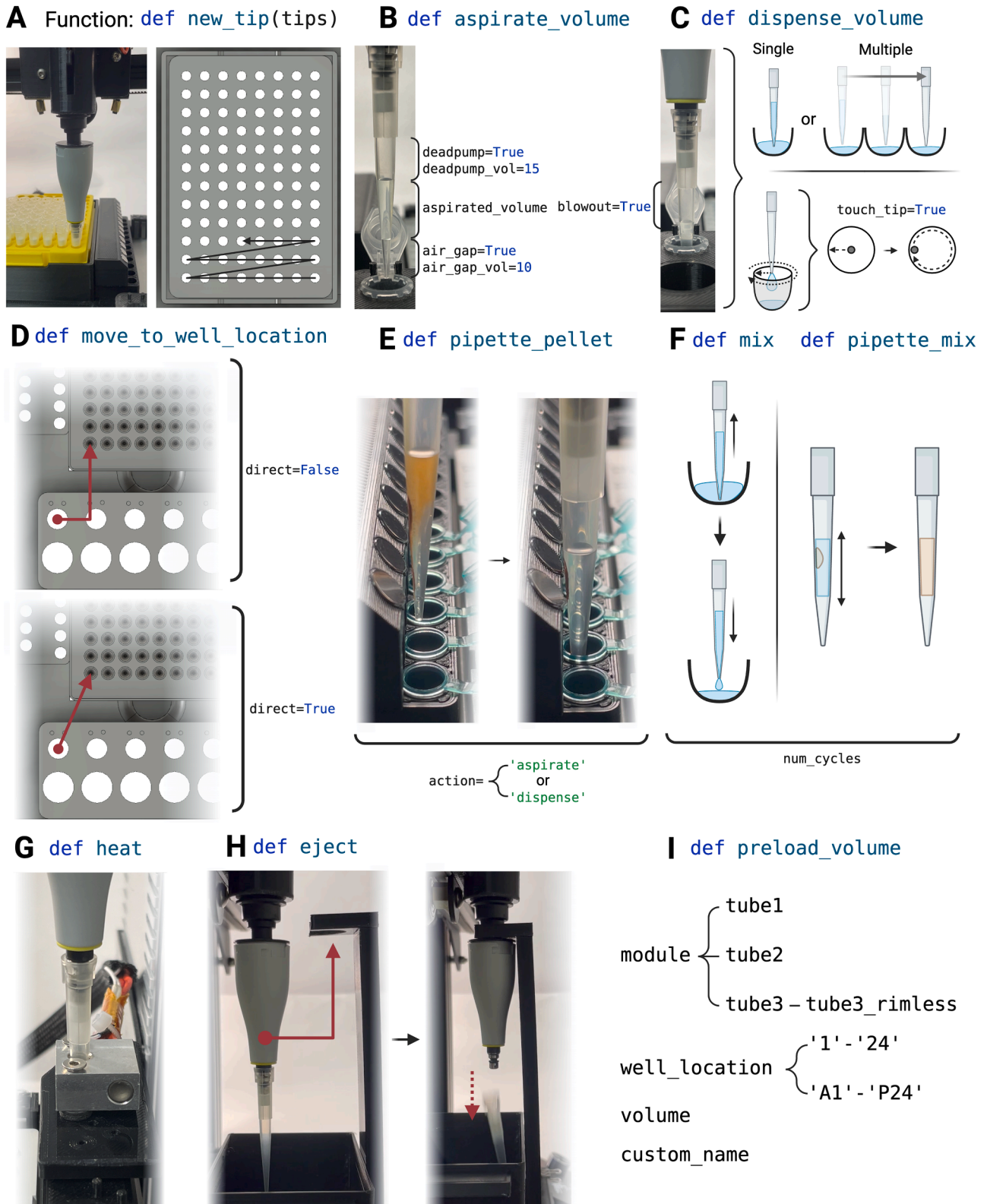


Fig. 5. Select functions and notable parameters in the Python program for generating G-code. Functions for: A) Getting a new tip, B) Aspirating select volume with options to specify a dead pump volume and an air gap volume after aspiration, C) Dispensing select volume for single or multiple dispenses from same aspirate, and touching the rim of the tube/well after dispensing to make sure all volume is left, D) Moving the pipet to a new location either indirectly or directly, E) Pelletting magnetic beads at the side of the pipet, F) Mixing select volume by aspirating and dispensing above the liquid surface, and mixing inside the pipet tips to dislodge magnetic beads, G) Heating pipet tip with the repurposed heater module, H) Ejecting pipet tip at the ejection site, and I) Specifying reagents included in the protocol for easy referencing.

bottom Z-axis coordinate (distance from the bed surface to the bottom of the well), safe Z-axis coordinates (where the pipet tip is above the well location), the distance (in mm) from the corner of the plate to the first well in X and Y axis, volume-to-height volume equation, radius of the well, and plate name. Additionally, the G1 code controlled the movement of the motors and was the basis of the PALH system's unique functions and other corresponding codes. For example, the defined G28 function homes all the motors with a set steps-to-unit for the motors and maximum feed rate where the homing is done sequentially as X-axis, Y-axis, and Z-axis motors; this is crucial to prevent crashes.

Before the first run after installation, the bed leveling needed to be calibrated using the *Z_tip_calibration.gcode* file, where the pipet component with a tip is positioned at the bed's four corners (Figure S4). Other calibration test scripts, *plate_384_biorad_cal_test.gcode* and *tube_only_calibration_test.gcode* were designed to make sure the pipet was moving correctly to the tube and the four corner wells of the 384-well plate locations sequentially. Once set up, a function such as *new_tip* was used to get new tips in sequential order starting from the right bottom of the pipet box (Fig. 5A). Whereas the *eject* function ejected the pipet tip by applying force directly to the pipet cone head from the tip ejector trigger (Fig. 5H).

The *aspirate_volume* function abridged aspiration of volume using the pipet with the option to specify if a dead pump (the aspiration of air volume before aspirating liquid) or an air gap (the aspiration of air volume after aspirating liquid) is utilized. The dead pump was an important feature for ensuring all the volume was completely dispensed with a blowout (a feature for fast dispensing the dead pump air volume after displacing liquid). The air gap, on the other hand, was an important feature for preventing any leftover liquid drip after aspirating liquid. It is worth noting that viscous liquids are more difficult to handle and often require slower aspiration speed, which can be adjusted as a parameter accordingly in the function. Low-surface tension fluids, particularly those with high vapor pressure, also pose a significant challenge during pipetting as they tend to drip from pipette tips during volume displacement [51]. This dripping can lead to inaccurate volume measurements and potential sample loss, compromising experimental precision and reproducibility. Currently, the PALH system does not have a function to automatically compensate for handling such high vapor-pressure reagents without a potential dripping effect. The aspirated volume was dispensed at the new well location using the function *dispense_volume*, where the liquid can be dispensed directly into a single destination or multiple destination locations (Fig. 5C). In a case where the air gap is not used, or the blowout is not sufficient, a touch tip feature can be used to ensure the liquid droplet at the pipet tip end can be dislodged at the inner surface of the well or tube by moving the pipet tip to the side of the well rim opening and circling around.

The *move_to_well_location* function dictated the movement of X and Y-axis motors, and it is referenced throughout the other movement functions (Fig. 5D). Depending on the workflow, indirect (move the X-axis first, then the Y-axis) or direct movements were used, and a safe distance of the Z-axis was also specified to ensure the pipet incorporated a safe height during movement. When handling magnetic beads, the *pipette_pellet* function was used to move the pipet tip close to the magnetic disk holder, where the Z-axis motor moved the pipet tip from the end of the tip to an ideal pelleting height (for example, when handling 100 μ L beads, a 12.2 mm Z value was used). The optimal z-coordinate height for pelleting magnetic beads was determined by positioning the pipet tip in close proximity to the bottom of the 0.2 mL tube while carefully avoiding suction effects. The magnetic disk assembly, set at a 45-degree angle relative to the tube holder, facilitated bead pelleting above the module. The pelleting height is specific to the 200 μ L pipet configuration. Hence any alterations in pipet dimensions or magnetic strength/size would necessitate recalibration of this parameter to maintain optimal performance.

A dispense action can be incorporated to dispense any supernatant, while the aspirate action can aspirate any wash or resuspension volume

during the pelleting function (Fig. 5E). Two different mixing functions were constructed: *mix* and *pipette_mix* functions. The *mix* function repeatedly aspirated volume at the bottom of the well and dispensed at the surface of any leftover liquid, while the *pipette_mix* function mixed the aspirated volume inside the pipet tip often to dislodge any pelleted magnetic beads (Fig. 5F). For heating the magnetic beads from outside the pipet tips, the *heat* function moves the pipet to a specified heating height for a selected temperature and duration (Fig. 5G). To simplify protocol scripting, the *preload_volume* function was created where the location and volume of the reagent are specified at the start of the script with a custom name and referenced back through the protocol.

2.3. Applications and performance

The efficacy of the PALH system in executing common life science assays for diagnostic workflows was evaluated here. Key methods such as serial dilution, reagent mixing, and magnetic purification were the focus of these studies. Specifically, workflows involving amplifying genomic DNA (gDNA) or vectors, isolation of amplicons, and purification of nucleic acid from crude samples were compared to traditional manual benchtop workflows.

One prevalent diagnostic assay is the detection of target genes using quantitative real-time polymerase chain reaction (qPCR) with target-specific primers. This study compared the quantitative cycle (Cq) values—indicating the number of cycles required for the relative fluorescence signal to surpass the background fluorescence—during the amplification of the housekeeping reference gene *YWHAZ*. The *YWHAZ* gene was chosen as it is commonly used for normalizing gene expression in qPCR workflows due to its abundance and stability [53]. Lower Cq values correspond to a higher initial quantity of the target nucleic acid, while higher Cq values indicate a lower initial quantity.

When using pure human gDNA as an input sample at 1:10 serial dilution ranging from 1 pg to 100 ng, a clear inverse linear relationship between sample concentration and Cq values was observed, indicating successful serial dilution in both manually prepared and PALH set-up plates (Fig. 6A). The comparison revealed that the PALH and manual qPCR preparation methods yielded comparable Cq values across various input sample quantities ($n = 3$ per condition) with no significant statistical differences. Notably, one of the manually prepared non-template control (NTC) reactions showed amplification, possibly due to random contamination or primer dimer formation [54]. This comparison highlights the PALH's capability to perform generic serial dilution workflows, reagent mixing, and 384-well plate loading. The *YWHAZ_qPCR_protocol.gcode* script was utilized to prepare the qPCR plates on the PALH [43].

End-point PCR sample preparation and nucleic acid purification for downstream processing are integral to many assay workflows, such as library preparation for next-generation sequencing (NGS) [55]. To simulate similar workflows, an end-point PCR was conducted using a pUC19 vector to amplify 204 bp and 406 bp amplicons. The amplified DNA was purified using solid-phase reversible immobilization (SPRI) magnetic beads, where DNA was isolated and washed with diluted ethanol to remove impurities before resuspending the magnetic beads in nuclease-free water to elute off the amplicons [56]. The sample preparation process for the amplification process was similar to how the qPCR plate was prepared. On the other hand, the magnetic purification process was performed using the novel pipet-based extraction process on the PALH, where magnetic beads are pelleted inside the pipet tips during supernatant removal and washing before the beads are resuspended in an elution buffer to elute off the captured amplicons.

A capillary-based gel electrophoresis comparison of PALH and manual purification methods of PCR products showed similar size traces at the expected 204 bp and 406 bp length. However, the PALH samples exhibited higher relative fluorescence for both amplicons compared to the manual processing method (Fig. 6B). This may be attributed to the pipet's ability to preserve the magnetic pellets during washing with

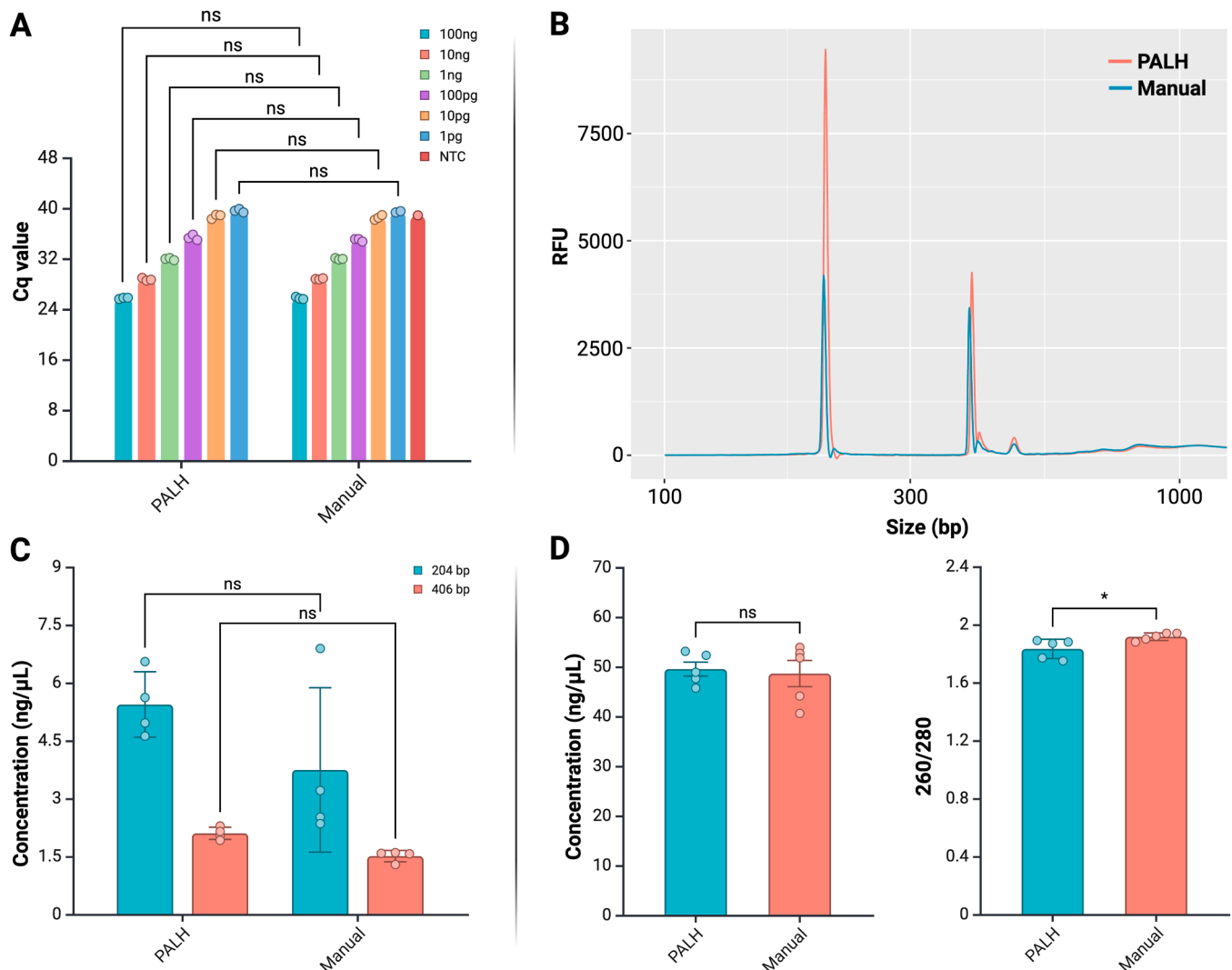


Fig. 6. Performance of the PALH and manually prepared sample processing in life science assay workflows. **A)** Comparison of YWHAZ gene qPCR amplification rate across serially diluted purified gDNA samples prepared using different loading methods ($n = 3$). **B)** Electropherogram tracing comparison of amplified and magnetically purified pUC19 samples at 204 bp and 406 bp. **C)** The final concentration (ng/ μ L) of the two different purified amplicons ($n = 4$). **D)** gDNA magnetic bead-based extraction from whole blood, comparing the final concentration (ng/ μ L) of gDNA and the purity of the purified sample using a 260/280 ratio ($n = 5$). The 'ns' signifies statistically not significant where $P > 0.05$, and '*' signifies statistical significance where $P \leq 0.05$. The error bars in the graph represents the standard deviation.

minimum disturbance and not elute off any amplicons before the elution step. The final concentration of purified amplicons was not significantly different between the PALH and manual methods, with values of 5.43 ± 0.85 ng/ μ L and 2.10 ± 0.16 ng/ μ L for the 204 bp amplicon and 3.74 ± 2.13 ng/ μ L and 1.51 ± 0.15 ng/ μ L for the 406 bp amplicon, respectively ($n = 5$, Fig. 6C; note the variation values ($\pm$) indicate the Standard Deviation). During isolation and washing, it was crucial to avoid depelleting the magnetic beads to maintain a high yield. This was achieved by keeping the pipet tips containing the pelleted magnetic beads in close proximity to the magnet disks at all times. The automation script implemented this protocol by programming the pipet tips to aspirate wash buffer from a specific well and dispense the waste back into the same well, ensuring the beads remained consistently magnetized. To ensure complete evaporation of any remaining wash buffer—which can decrease the efficiency of any downstream processing—the pipet tip with pelleted beads was heated using the heating module, and the wash buffer residue was removed via evaporation [57,58]. This experiment demonstrated the extension of the PALH's capability to isolate DNA magnetically. The *pUC19_amplification.gcode* and *pUC19_cleanup.gcode* scripts were used for PCR and cleanup workflows [43].

Magnetic bead-based extraction of gDNA from crude samples, such as peripheral whole blood, is a common workflow in diagnostic automation systems [59,60]. To extract gDNA from peripheral blood, blood cells were lysed to release gDNA using a lysis buffer that breaks down cell membranes and nuclear envelopes [61]. Similar to the pUC19 workflow, naked gDNAs were captured by magnetic beads under binding conditions, followed by multiple washes to remove contaminants [62]. However, unlike the pUC19 workflow, the magnetic beads were resuspended from their pelleted form by washing them multiple times to remove any lysate, protein, lipid, and other contaminants with the wash buffer [62]. The pelleted magnetic beads inside the pipet tip were dried using a heating module, and gDNA was eluted with an elution buffer without any magnetic field presence. The drying step was important to ensure that the wash buffer did not carry over to the elution buffer and contaminate any downstream steps, such as amplification.

Successful extraction is assessed by the quantity and quality of the purified gDNA. Here, the final yield of extracted and purified gDNA from peripheral blood using PALH and manual methods showed no significant difference in DNA concentration, with values of 49.52 ± 3.13 ng/ μ L and 48.62 ± 5.9 ng/ μ L, respectively, from 100 μ L of peripheral whole

blood when eluted in 33 μL . The purity of extracted DNA is determined by the 260/280 ratio, where the absorbances of a nucleic acid sample at two wavelengths—260 nm and 280 nm—are compared. A ratio of around 1.8 indicates a “pure” extracted DNA [63]. The PALH system yielded a 260/280 ratio of 1.83 ± 0.07 , while the manual method yielded a ratio of 1.92 ± 0.03 , suggesting a statistically significant difference favoring the PALH method for DNA purity ($n = 5$; Fig. 6D). While there are some discrepancies in the wavelength ratio between manual and PALH system extraction methods, they are still within the acceptable range of 1.8–2.0. However, this discrepancy may be due to the mixing mechanism of the PALH and manual methods or the incorporation of RNA or other contaminants, such as proteins or phenol with lower wavelength peak absorbance [63]. The *magnetic_extraction_w_heater.gcode* was used to prepare the PCR and the cleanup workflows [43].

These evaluations of the PALH system demonstrate its capability to perform a select range of common life science assays for diagnostic workflows. The automated system successfully executed key methods used in life science methods, showing comparable performance to traditional manual benchtop methods. The PALH system effectively prepared qPCR, processed end-point PCR samples, and extracted gDNA from crude blood samples. Other limitations include the limited capacity to handle widely different ranges of volumes, dependent on the pipet module, which is not easily interchangeable. The liquid handler's capacity to hold only a single channel restricts its scalability and high throughput capacity.

Additionally, the current iteration of the PALH system needs to demonstrate a significant time-efficiency advantage over manual processes, particularly when considering that skilled technicians using multichannel pipets can process up to eight samples simultaneously. In fact, the PALH system may increase processing time across various life-science workflows due to its limited, slow velocity of gantry movement and volume displacement. Consequently, the PALH system's utility for expediting processes is limited unless it is specifically employed for pooled sample manipulation. Hence, the system's primary benefits may lie in areas other than time efficiency, such as increased precision or reduced human error, rather than in accelerating sample processing. However, with the timely advancement of single-channel high multiplexing workflows, the PALH system may become clinically effective [64].

Unlike typical commercial automated liquid handlers that employ a stationary bed with only the pipetting head moving in the XYZ axis, the PALH system utilizes a separate Y-axis movement for the bed itself while the pipetting head moves in the X and Z axis. This introduces a theoretical risk of liquid spillage if the bed moves at excessive speeds or experiences rapid acceleration or deceleration. To mitigate such risk, maximum feedrate controls are used at the start of every protocol where the X, Y, and Z axis have velocities of 6000, 1000, and 30 mm/s, respectively, while the pipet volume displacement feed rate is 20 $\mu\text{L}/\text{s}$ (as reported in the *gcode_generator_v1.2.py* file) [43]. The feedrates used were different depending on the specific steps in the script.

Another significant limitation of the study in assessing the PALH's performance is the inherent variability in manual pipetting, where the performances are compared. Pipetting is highly user-dependent, which can ultimately skew the comparison in this study. Consequently, the reported manual data may not provide an optimal benchmark for evaluating the PALH's efficacy; hence, comparison studies between the PALH and other established automated liquid handling systems would offer a more standardized and reliable basis for the assessment. Finally, the low cost of the printer is mainly due to the low-quality parts of the instrument, which may lead to accelerated wear and potentially compromise the system's durability and precision over time. This again necessitates a thorough investigation into the PALH's long-term performance and the assessment of the system's technological readiness level (TRL), including studies on the frequency of required calibrations and the overall lifespan of the system compared to more robust,

commercial automated liquid handlers. With the validation of the PALH system in the lab, the technology may be at TRL of 4 and would need further investigation to improve its readiness.

Overall, the PALH system shows great promise as an affordable and low-throughput automated alternative to manual processes, offering consistent results across various assays while potentially reducing human error. This study lays the groundwork for further development and application of the PALH system in diverse life science and diagnostic settings, paving the way for more efficient and reliable automated laboratory processes.

3. Materials and methods

3.1. Component printing & platform setup

The Ender-3 3D printer (Creality, Shenzhen, China) with mainboard V4.2.7 was assembled using the manufacturer's instructions. The open-sourced Marlin firmware V2.1.2.2 for the printer was modified using Visual Studio Code (Microsoft, Seattle, WA) and built using the Platform IO extension [65,66]. The updated *firmware-printer.bin* was installed on the printer using an Ender-3 microSD card. The modules for the PALH system (Table S1) were designed using CAD design software SOLIDWORKS (Dassault Systèmes SOLIDWORKS, Waltham, MA) and Fusion 360 (Autodesk, San Francisco, CA). The G-code files for the 3D models were generated using the Creality Slicer V4.8.2 with a modified bed size of 238×238 mm [2]. The designed modules were printed using the Ender 1.75mm PLA 3D Printing Filament (Creality). During printing, four binder clips were placed at the four corners of the printing bed, as shown in Figure S2.

Once all models were printed, the 3D printer was reconfigured to modify the system to a PALH. The E1 ClipTip Tip cone assembly (Pipet Supplies, Mansfield, TX) was assembled following the manufacturer's instructions and fitted into the 3D-printed parts. The stepper motor linear actuator (Haydon Kerk Pittman, Waterbury, CT) and the micro limit switch (MXRS-Amazon, Seattle, WA) follow the Fig. 2B assembly. The deck base and the ejector modules were positioned following the Fig. 2A assembly using the fitting screws (McMaster-Carr, Elmhurst, IL), as shown in Table 2. The assembled components were then installed following Fig. 7A–B. The wiring of the pipet motor and the mainboard were soldered following Fig. 7C. The optional components, the magnet holder module, and the heater module, were assembled following Fig. 2C–D. The magnetic holder used 12x D42-Neodymium disc magnets (K&J Magnetics, Pipersville, PA). The heater module reused the extruder's nozzle heater block without the insulator and was installed inversely on the heater block holder. The Marlin firmware was again updated with corresponding changes (all modifications viewable on Zenodo), and the built *firmware_PALH_v1.3.1.1.bin* was installed on the printer using an Ender-3 microSD card [43].

The potentiometer controlling the voltage on the E stepper port was adjusted with a screwdriver to enable the pipet motor without overheating, as shown in Fig. 8. The reconfigured connection for the PALH was also followed using Fig. 8. The G-code generator Python code was constructed internally and tested using the *Printrun* host software suite [67,68].

3.2. Reaction setup

The qPCR workflow used human male genomic DNA (Promega, Madison, WI) as the input template with varying quantities to detect the YWHAZ target gene [70]. The qPCR amplification incorporated 250 nM of probe 5'-/5Cy5/ ACT GTC AAG /TAO/ TTG TCT CTC AGT AAT TGC /31AbRQSp/-3', 500 nM of forward primer 5'-AAC ATA ACC TC TTC AAC CAC AT-3', 500 nM of reverse primer 5'-GAA GAG TCA TAC AAA GAC AGC A-3', and 1X PrimeTime Gene Expression Master Mix in a 20 μL reaction. The primers and the master mix were ordered from IDT (Integrated DNA Technologies, Coralville, IA) and followed the

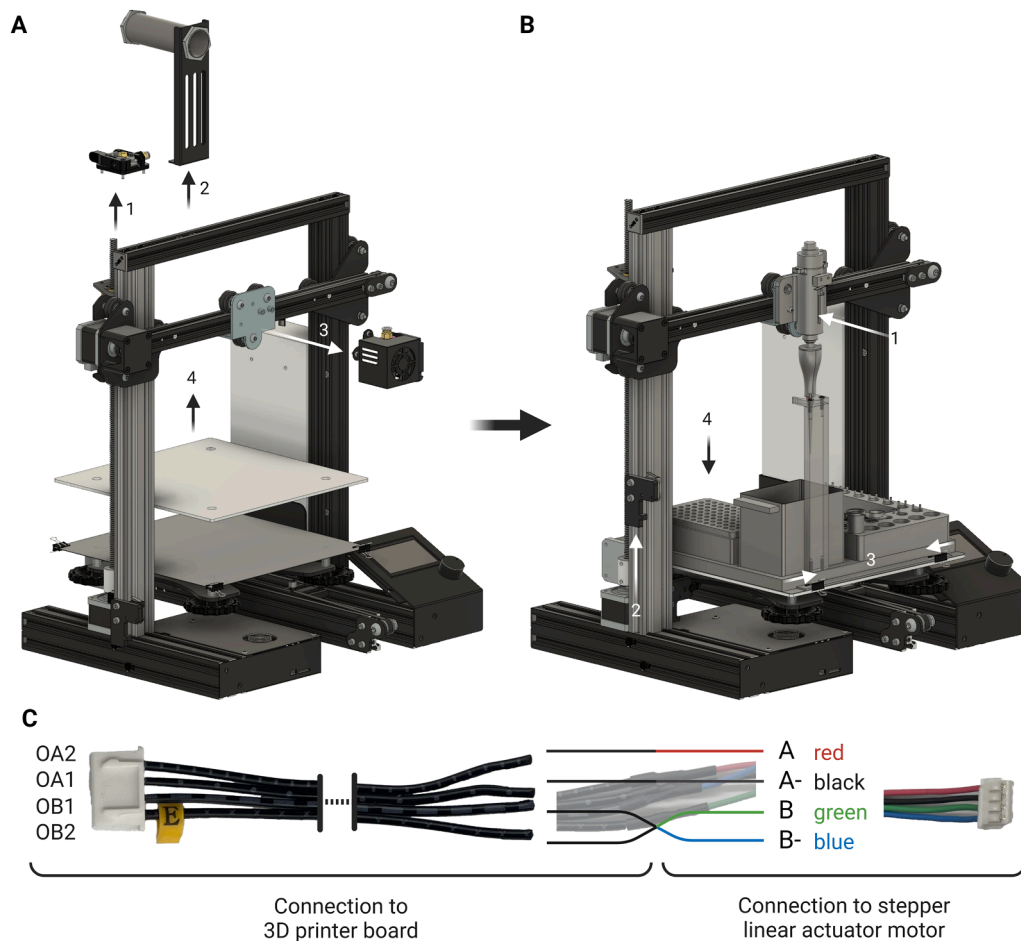


Fig. 7. The disassembly and configuration of converting the 3D printer to the PALH system. **A)** The replacement of the extruder module: 1. Removal of the filament guide (optional), 2. Removal of the filament spool (optional), 3. Removal of the extruder nozzle heater and casing, and 4. Removal of hotbed. **B)** Installment of PALH components: 1. Installation of the assembled pipet component at the same securing threads, 2. The displacement of the Z-axis homing limit switch for the distance of 102.22 mm between the base of the Z-axis limit switch holder and the base of the 3d printer rail, 3. The displacement of the four binder clips to clamp the deck base to the bed of the 3D printer, and 4. The installment deck base component onto the 3D printer bed. **C)** The junction configuration for reconnecting the extruder wiring to the pipet stepper linear motor to control the pipet volume displacement function.

manufacturer's instructions for the cycling program. The amplification setup was loaded into Hard-Shell 384-well PCR plates (BioRad, Hercules, CA), and samples were thermal cycled, with fluorescence measured using the CFX384 Touch Real-Time PCR Detection System (BioRad). Amplification values were statistically compared using two-way ANOVA with the Tukey multiple comparisons test.

The end-point PCR and purification workflow used 10 ng of pUC19 vector (New England Biolabs, Ipswich, MA) as the input template, 200 nM forward primer 5'-AGG AGA AAA TAC CGC ATC AG-3' for the 406 bp amplicon, 200 nM forward primer 5'-GAT CCT CTA GAG TCG ACC TG-3' for the 206 bp amplicon, 200 nM universal reverse primer 5'-CAG GTT TCC CGA CTG GAA AG-3' (Integrated DNA Technologies, Coralville, IA), and 1X Hot Start Taq Master Mix 50 μ L reaction (New England Biolabs) in 200 μ L reaction tubes (Eppendorf, Hamburg, Germany). The amplification followed a thermal cycle of 95 $^{\circ}$ C for 60 s as the initial denaturation, followed by 30 cycles of 95 $^{\circ}$ C for 15 s, 50 $^{\circ}$ C for 30 s, 68 $^{\circ}$ C for 30 s, and ending with the final extension of 62 $^{\circ}$ C for 300 s using Mastercycler nexus X2 (Eppendorf). The amplicons were purified using AMPure XP magnetic beads (Beckman Coulter, Brea, CA) at 1.5X sample volume and incubated for 5 min before applying magnets to the tip and removing the supernatant. While still on the magnetic stand, without disturbing the magnetized pellet, 200 μ L of 70% isopropyl alcohol (IPA; Sigma Aldrich, St. Louis, MO) was added and incubated for 30 s before removing any remaining supernatant. The IPA washing was done twice

before removing the sample from the magnetic stand and air dried for 2 min. 33 μ L of nuclease-free water (Integrated DNA Technologies) was used to resuspend the pellet completely and incubated for 2 min before transferring 30 μ L of the eluted sample to a new container. The amplified products were analyzed using the Agilent Bioanalyzer 2100 electrophoresis system with a High Sensitivity DNA kit (Agilent, Santa Clara, CA) following the manufacturer's instructions.

The gDNA extraction workflow used fresh human peripheral whole blood (Research Blood Components, Watertown, MA) in EDTA anticoagulation vacutainers. The blood samples were frozen at -20° C upon arrival and were used after two weeks. 100 μ L of blood sample was used as the input for the chemagic Body Fluid 1k Kit H96 (Revvity chemagen, Baesweiler, Germany), and the gDNA extraction was completed following the manufacturer's instructions. The final extracted samples were analyzed using the NanoDrop 1000 Spectrophotometer (Thermo-Fisher Scientific, Waltham, MA). The extracted DNA quantity (ng/ μ L) and quality (A260/280) were statistically compared using an unpaired *t*-test. Amplified Cq values were statistically compared using two-way ANOVA with the Tukey multiple comparisons test.

To support the workflows, two plate identifications—96-well and 384-well plates (BioRad) were created in the PALH *gcode_generator_v1.2.py* code. The corresponding reagent locations and protocol steps for the PALH system can all be found in the *Zenodo* public repository [43]. All

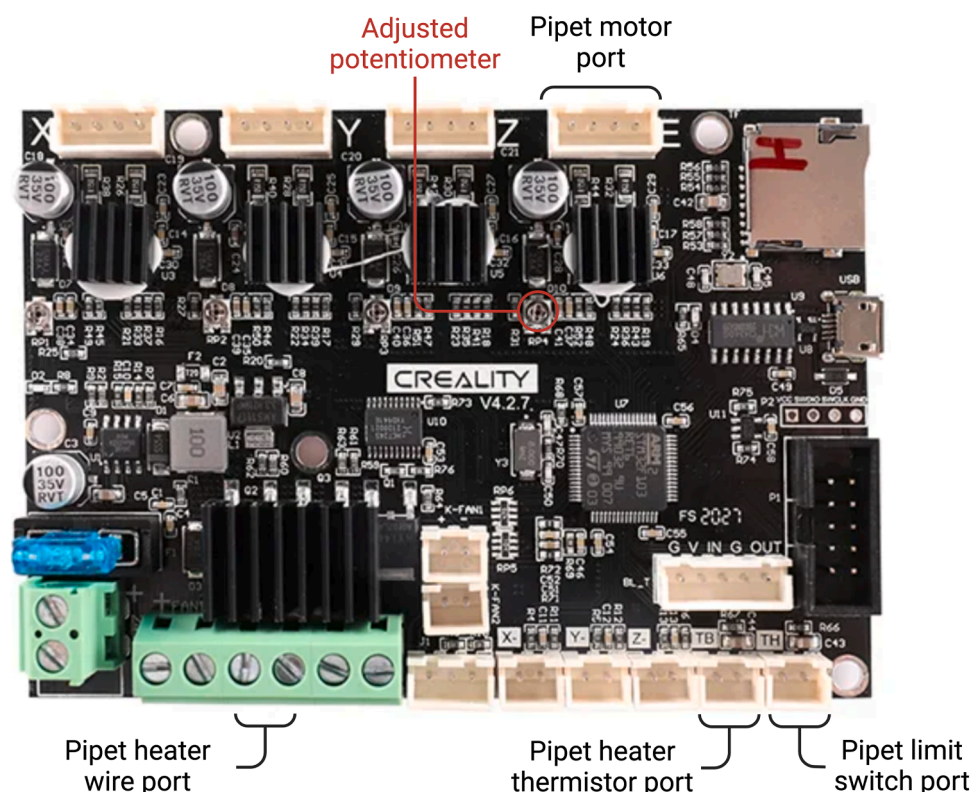


Fig. 8. Illustration of reconfigured Creality V4.2.7 mainboard port connections (board image from Creality.com) [69].

figures and illustrations were post-processed or made using Biorender.com (BioRender, Toronto, ON). At the same time, the statistical comparisons were analyzed using R version 4.2.2 (R Foundation for Statistical Computing, Vienna, Austria) in BioRender Graph Illustrator.

4. Conclusion

This study showcased the feasibility of repurposing a 3D printer into a personal automated liquid handler (PALH) system, which provides a cost-effective, customizable, and open-source alternative to traditional automated liquid handling systems for life science laboratories. With the capacity to 3D print module components, the PALH system significantly reduced the financial barrier to entry for laboratory automation at a total base cost of approximately \$400.55.

The performance evaluated by the PALH system across various applications yielded promising results. In qPCR sample preparation, the PALH system demonstrated comparable performance to manual methods, with consistent Cq values obtained across different input sample quantities. For end-point PCR sample preparation and nucleic acid purification, the PALH system showed proficiency in maintaining sample integrity throughout complex workflows such as magnetic bead-based extraction using an innovative pipet-based extraction method. In gDNA extraction from whole blood samples, the PALH system demonstrated efficiency comparable to manual methods in terms of DNA yield. However, a slight discrepancy in DNA purity was observed, with PALH processed samples showing marginally better 260/280 ratios (1.92 ± 0.03 vs. 1.83 ± 0.07 for PALH). This finding highlights an area for potential optimization in the PALH system, particularly in the washing steps of magnetic bead-based purification protocols.

The PALH system's ability to perform various liquid handling tasks demonstrates its versatility and potential value for various applications in molecular biology and diagnostics. Its open-source nature and customizability may appeal to further development, allowing researchers to adapt and optimize the platform for specific experimental needs. As the

scientific community continues to embrace open-source technologies, systems like PALH have the potential to accelerate research, improve reproducibility, and enable more laboratories to benefit from automated workflows. Future developments and collaborations building upon this foundation could further enhance the capabilities and applications of such systems, ultimately contributing to more efficient and accessible scientific research.

Data Availability

The corresponding source codes for 3D printing modules, creating a protocol, and building firmware can be found at *Zenodo* public repositories [43].

The sequence for the YWHAZ gene (range: 100,916,523–100,953,382 of the Homo sapiens chromosome 8, GRCh38.p14) can be accessed at NCBI as Reference Sequence: NC_000008.11 [70]. The cloning vector pUC19 complete sequence can be accessed at GenBank: M77789.2 [71].

Ethical statement

The peripheral whole blood samples used in this study were acquired from Research Blood Components following the American Association of Blood Banks guidelines for drawing donors [72,73]. The samples were anonymized and obtained from patients who provided written informed consent. All personally identifiable information was excluded to ensure donor confidentiality during the study.

Competing interests

The authors declare no conflict of interest regarding this work. AT is a paid scientific advisor/consultant and lecturer for Revvity Inc.

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CRediT authorship contribution statement

Dulguunnaran Naranbat: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Benjamin Phelps:** Writing – review & editing, Validation, Software, Methodology, Investigation, Conceptualization. **John Murphy:** Writing – review & editing, Validation, Resources, Methodology, Investigation, Conceptualization. **Anubhav Tripathi:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Anubhav Tripathi reports financial support was provided by Revvity Inc. Dulguunnaran Naranbat reports financial support was provided by Revvity Inc. John Murphy reports financial support was provided by Revvity Inc. Anubhav Tripathi reports a relationship with Revvity Inc that includes: funding grants and speaking and lecture fees. The authors declare no conflict of interest regarding this work. AT is a paid scientific advisor/consultant and lecturer for Revvity Inc. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.slast.2024.100239](https://doi.org/10.1016/j.slast.2024.100239).

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